Part I - Short Answer Questions

Q1) A Database Management System (DBMS) is used to organize and manage data efficiently. It has features like security, backup, recovery, and multi-user access. In contrast, a file system stores data in separate files and does not provide these facilities. A DBMS reduces duplicate data and helps keep information accurate and consistent.
Scenario: A college management system is better implemented using a DBMS because different departments need to access and update student information simultaneously. The DBMS ensures that the data remains secure and consistent.

Q2) A strong entity can be identified by its own primary key and does not depend on any other entity. A weak entity cannot be uniquely identified by its attributes alone and depends on a strong entity.

The key of a weak entity is formed by combining:

- The primary key of the strong entity
- The partial key of the weak entity

The relationship connecting them is known as an identifying relationship

Q3) Phone number should be represented as a multivalued attribute because a student may have more than one phone number. A simple attribute stores only one value, while a composite attribute can be divided into smaller parts. Therefore, phone numbers are best modeled as a multivalued attribute.

Q4) A super key is any attribute or combination of attributes that can uniquely identify a record in a table. A candidate key is a super key with no extra attributes. If any attribute is removed, it can no longer uniquely identify the record. Therefore, every candidate key is a super key, but not every super key is a candidate key.

For the relation Student(roll_no, email, name, branch), since roll_no of a student will always be unique and assuming email is also unique

Candidate Keys:

- roll_no
- email

Examples of super keys that are not candidate keys:

- (roll_no, name)
- (roll_no, branch)
- (email, name)
- (email, branch)

These are super keys because they uniquely identify students, but they contain extra attributes.

Q5)

- The constraint that is violated here is referential integrity constraint.
- It ensures that every student_id in the Enrollment table refers to a student that exists in the Student table.
- The insert operation should be rejected (or fail) because the student_id does not exist in the Student table.

Q6)

- A separate junction table must be created.
- The primary key of the junction table consists of the primary keys of the participating entities combined together.
- A single column cannot store all these connections properly, because one student can be linked to many courses and one course can be linked to many students.

Part II - Long Answer Questions

Part A – Entity and Relationship Identification

Entities and Key Attributes:

1. Customer (customer_id)
2. Restaurant (restaurant_id)
3. Order (order_id)
4. MenuItem (item_id)

Relationships:

1. Customer – Order

   Cardinality - 1:N because one customer can place many orders, but each order belongs to only one customer.

2. Restaurant – Order

   Cardinality - 1:N because a restaurant can receive many orders, but each order is associated with only one restaurant.

3. Restaurant – MenuItem

   Cardinality - 1:N because a restaurant can offer multiple menu items, but each menu item belongs to one restaurant.

4. Order – MenuItem

    Cardinality - M:N because an order may contain several menu items, and a menu item can be included in many orders.

Part B – Attribute Classification

Multivalued Attribute:

Customer phone number because a customer may have more than one phone number.

Composite Attribute:

Customer address because address can be divided into smaller parts such as house number, street, city, state, and PIN code.

Part C – Relational Schema Design

Since the relationship between Order and MenuItem is M:N, a junction table is required.

Schema:

OrderMenuItem(order_id, item_id, quantity)

Primary Key - (order_id, item_id)

Foreign Keys:

- order_id
- item_id

Additional Attribute - quantity

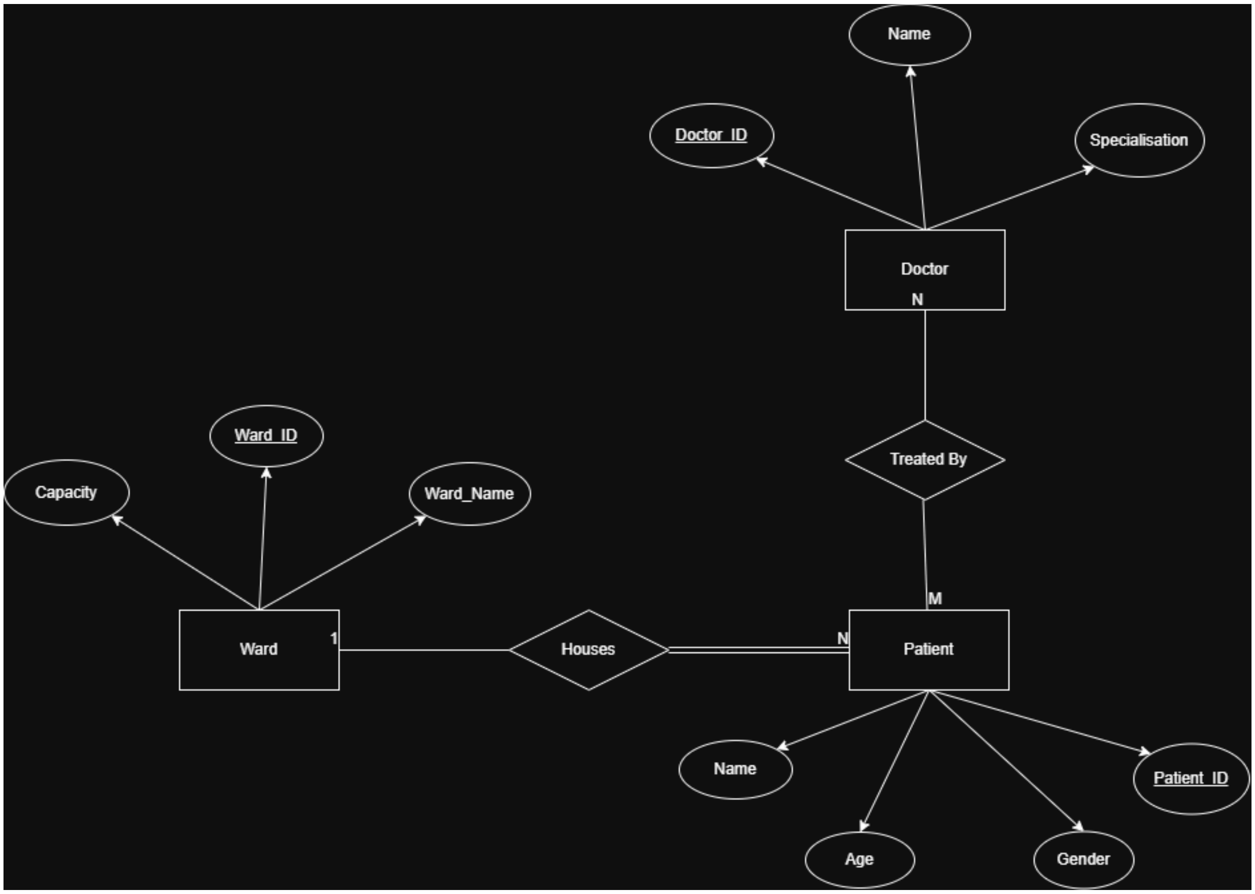
Final Schema:

OrderMenuItem(order_id, item_id, quantity)

- Primary Key: (order_id, item_id)
- Foreign Keys: order_id, item_id
- Additional Attribute: quantity

Part III - ER Diagram Design

Question 1: Hospital ER Diagram



Question 2: Design Justification

The relationship between doctors and patients is many-to-many because a doctor can treat several patients, and a patient can receive treatment from multiple doctors. To represent this relationship in a database, an associative entity called Treatment can be created. This entity connects doctors and patients and may also store information such as treatment date or diagnosis. The composite primary key of the Treatment entity would be made up of Doctor_ID and Patient_ID.

In the Ward–Patient relationship, the patient has total participation because every patient must be admitted to a ward. A ward, however, has partial participation because it can exist even when no patients are currently assigned to it. In the Doctor–Patient relationship, both doctor and patient have partial participation because a doctor may temporarily have no patients, and a patient may not yet be assigned to a doctor.

One attribute design choice is the patient's address. Address is modeled as a composite attribute because it can be divided into smaller parts such as house number, street, city, state, and PIN code. Storing these parts separately makes the data easier to organize, search, and manage.